

Homework 3

Department: Computer Science and Engineering

Invigilator's Signature

Total No. of Pages

Mid Semester

End Semester

Make-up

Suppl. Exam

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Programme: BTech

(BTech / BDes / MSc/MTech/MDes/MA/PhD etc)

Course No.: EE5750

Course Title: Quantum Error Correction

PART: •

Question No.	1	2	3	4	5	6	7	8	9	10	Total
MARKS											

Homework-3

1. $\Rightarrow$ Hilbert space $C^d \Rightarrow \{|\psi_i\rangle : i=0 \text{ to } d-1\}$ be any orthonormal basis (need not be standard basis)

$$\Rightarrow P_i = |\psi_i\rangle\langle\psi_i|$$

(a) $\Rightarrow P_i^2 = |\psi_i\rangle\langle\psi_i||\psi_i\rangle\langle\psi_i| = |\psi_i\rangle\langle\psi_i||\psi_i\rangle\langle\psi_i|$
 $= |\psi_i\rangle\langle\psi_i|$ [$\because \langle\psi_i|\psi_i\rangle = 1$ as orthonormal basis]

$$\boxed{P_i^2 = P_i}$$

$$\Rightarrow P_i^\dagger = (|\psi_i\rangle\langle\psi_i|)^\dagger = |\psi_i\rangle\langle\psi_i| \Rightarrow \boxed{P_i^\dagger = P_i}$$

$\Rightarrow$ P_i is a projection operator

(b) $\Rightarrow \forall |\phi\rangle \in C^d, P_i|\phi\rangle = \underbrace{|\psi_i\rangle\langle\psi_i|}_{\text{scalar multiple of } |\psi_i\rangle}|\phi\rangle$

Column space of $P_i = \boxed{\text{Col}(P_i) = \text{span}\{|\psi_i\rangle\}}$

$$\text{Col}(P_i) = \{c|\psi_i\rangle \mid c \in \mathbb{C}\}$$

$\Rightarrow |\psi_i\rangle \neq 0$, $\text{Col}(P_i)$ is a 1-dimensional subspace

$$\boxed{\text{rank}(P_i) = 1}$$

(c) $\Rightarrow$ Given $i \neq j$
 $\Rightarrow P_i P_j = |\psi_i\rangle \langle \psi_i| |\psi_j\rangle \langle \psi_j| = |\psi_i\rangle (\langle \psi_i | \psi_j \rangle) \langle \psi_j|$
 $\Rightarrow$ If $i \neq j$, by orthonormality of basis, $\langle \psi_i | \psi_j \rangle = 0$
 $\Rightarrow P_i P_j = |\psi_i\rangle (0) \langle \psi_j| \Rightarrow \boxed{P_i P_j = 0 \text{ for } i \neq j}$

(d) $\Rightarrow$ Take any $|\phi\rangle \in \mathbb{C}^d$. As $\{|\psi_i\rangle\}$ is orthonormal basis,
 $|\phi\rangle = \sum_{j=0}^{d-1} c_j |\psi_j\rangle$ where $c_j = \langle \psi_j | \phi \rangle$

$\Rightarrow$ Apply $\sum_i P_i$ to above eqⁿ

$$\Rightarrow \sum_{i=0}^{d-1} P_i |\phi\rangle = \sum_{i=0}^{d-1} |\psi_i\rangle \langle \psi_i | \phi \rangle$$

$$= \sum_i |\psi_i\rangle \langle \psi_i | \left(\sum_j c_j |\psi_j\rangle \right)$$

$$= \sum_i \sum_j c_j |\psi_i\rangle \langle \psi_i | \psi_j \rangle = \sum_i \sum_j c_j \delta_{ij} |\psi_i\rangle$$

$$= \sum_i c_i |\psi_i\rangle = |\phi\rangle = \left(\sum_{i=0}^{d-1} P_i \right) |\phi\rangle$$

$$\Rightarrow \boxed{\sum_{i=0}^{d-1} P_i = I_{d \times d}} \rightarrow \text{Completeness condition.} \quad \hookrightarrow \text{identity operator}$$

$\Rightarrow$ When we choose $|\psi_i\rangle = |i\rangle$ i.e. when orthonormal basis used for measurement is standard basis, we say that we perform measurement in computational basis

2.

$\Rightarrow P$ and Q are projection operators $\Rightarrow PQ = 0$

$$\Rightarrow P^2 = P, Q^2 = Q, P^\dagger = P, Q^\dagger = Q$$

$$(a) \quad QP = Q^{\dagger} P^{\dagger} = (PQ)^{\dagger} = 0^{\dagger} = 0 \Rightarrow \boxed{QP = 0}$$

(b) $\Rightarrow |\psi\rangle$ is eigenvector of Q with eigenvalue 1

$$\Rightarrow Q|\psi\rangle = |\psi\rangle$$

$$\Rightarrow P|\psi\rangle = P(Q|\psi\rangle) = PQ|\psi\rangle = 0|\psi\rangle$$

$$\Rightarrow \boxed{P|\psi\rangle = 0}$$

(c) $\Rightarrow$ Let $R = P + Q$

$$\Rightarrow R^2 = (P+Q)^2 = P^2 + Q^2 + PQ + QP = P + Q + 0 + 0 \\ = P + Q \Rightarrow \boxed{R^2 = R}$$

$$\Rightarrow R^{\dagger} = (P+Q)^{\dagger} = P^{\dagger} + Q^{\dagger} = P + Q \Rightarrow \boxed{R^{\dagger} = R}$$

$\Rightarrow \boxed{P+Q \text{ is a projection operator}}$

$\Rightarrow$ We can show that $\text{rank}(P+Q) = \text{rank}(P) + \text{rank}(Q)$ and $P+Q$ projects onto subspace obtained as sum of column spaces of P and Q

4. $\Rightarrow$ Consider a classical measurement of classical states
 $\Rightarrow$ Consider a partition $S_1, \dots, S_L$ of $\{0, 1, \dots, d-1\}$ and distinct real no.s $\lambda_1, \lambda_2, \dots, \lambda_L$. The measurement device outputs λ_a if the classical RV X stored in classical register belongs to S_a i.e. $X \in S_a$

$$\Rightarrow \text{Operator } M = \sum_{a=1}^L \lambda_a P_a$$

$$\Rightarrow P_a = \sum_{j \in S_a} |j\rangle \langle j|$$

ρ is classical state

$$(a) \Rightarrow M\rho = \left(\sum_{\ell} \lambda_{\ell} \sum_{j \in S_{\ell}} |j\rangle\langle j| \right) \left(\sum_k P_k |k\rangle\langle k| \right)$$

$$= \sum_{\ell} \sum_{j \in S_{\ell}} \lambda_{\ell} P_j |j\rangle\langle j|$$

$$\Rightarrow \text{Tr}(M\rho) = \text{Tr} \left(\sum_{\ell} \sum_{j \in S_{\ell}} \lambda_{\ell} P_j |j\rangle\langle j| \right)$$

$$= \sum_{\ell} \lambda_{\ell} P(X \in S_{\ell})$$


 prob. of $X \in S_{\ell}$

$$\Rightarrow \boxed{\text{Tr}(M\rho) = \sum_{\ell} \lambda_{\ell} P(X \in S_{\ell})}$$

→ expected value of measurement outcome.

Device outputs λ_{ℓ} iff $X \in S_{\ell}$.

$$(b) \Rightarrow M^2 = \left(\sum_{\ell} \lambda_{\ell} P_{\ell} \right)^2 = \sum_{\ell} \lambda_{\ell}^2 P_{\ell}$$

$$[P_{\ell} P_{\ell'} = P_{\ell} \text{ if } \ell = \ell' \text{ and } 0 \text{ otherwise}]$$

$$\Rightarrow M^2\rho = \left(\sum_{\ell} \lambda_{\ell}^2 P_{\ell} \right) \left(\sum_k P_k |k\rangle\langle k| \right)$$

$$P_{\ell} = \sum_{j \in S_{\ell}} |j\rangle\langle j|$$

$$\Rightarrow M^2\rho = \sum_{\ell} \sum_{j \in S_{\ell}} \lambda_{\ell}^2 P_j |j\rangle\langle j|$$

$$\Rightarrow \text{Tr}(M^2\rho) = \sum_{\ell} \sum_{j \in S_{\ell}} \lambda_{\ell}^2 P_j$$

$$\boxed{\text{Tr}(M^2\rho) = \sum_{\ell} \lambda_{\ell}^2 P(X \in S_{\ell})}$$

→ second moment of measurement outcome

5. $\Rightarrow$ Pauli Z operator $\rightarrow Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

(a) $\Rightarrow |Z - \lambda I| = \begin{vmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \\ = \begin{vmatrix} 1-\lambda & 0 \\ 0 & -1-\lambda \end{vmatrix} \\ = (1-\lambda)(-1-\lambda) = -(1-\lambda)(1+\lambda) = \lambda^2 - 1$

$|Z - \lambda I| = 0 \rightarrow \lambda^2 = 1 \rightarrow \boxed{\lambda_1 = +1} \quad \boxed{\lambda_2 = -1}$
 $\underbrace{\hspace{10em}}$ Eigenvalues

$\Rightarrow Z \cancel{V_1} V_1 = \lambda_1 V_1$; V_1 is eigenvector corresponding to λ_1

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} V_{11} \\ V_{21} \end{bmatrix} = \begin{bmatrix} V_{11} \\ V_{21} \end{bmatrix} \rightarrow \begin{matrix} V_{11} = V_{11} \\ -V_{21} = V_{21} \end{matrix}$

$\boxed{V_1 = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}} \rightarrow \text{Eigenvector corresponding to } \lambda_1 = +1$

$\Rightarrow Z V_2 = \lambda_2 V_2 \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix} = \begin{bmatrix} -V_{12} \\ -V_{22} \end{bmatrix}$
 $\underbrace{\hspace{10em}}$ V_2

$V_{12} = -V_{12} ; -V_{22} = -V_{22}$

$\boxed{V_2 = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}} \rightarrow \text{Eigenvector corresponding to } \lambda_2 = -1$

(b) $\Rightarrow Z$ is used as an observable for a measurement

$\Rightarrow$ Measurement outcomes are Eigenvalues

$\boxed{\lambda_1 = +1} \quad \boxed{\lambda_2 = -1}$

$\Rightarrow$ Projection operators:

$P_1 = |V_1\rangle\langle V_1| = |0\rangle\langle 0| \rightarrow \boxed{P_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}$

$P_2 = |V_2\rangle\langle V_2| = |1\rangle\langle 1| \rightarrow \boxed{P_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}$

$$(c) \Rightarrow Z \otimes Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = -1, \lambda_4 = -1$ are eigenvalues
(1 with multiplicity 2, -1 with multiplicity 2)

$\Rightarrow$ Distinct measurement outcomes: $\boxed{+1 \text{ and } -1}$

$\Rightarrow$ Eigenvectors corresponding to eigenvalue 1
are $|00\rangle$ and $|11\rangle$

Projection operator:

$$P_1 = |00\rangle\langle 00| + |11\rangle\langle 11| =$$

$$\boxed{\lambda_1 = 1}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = P_1$$

$\Rightarrow$ Eigen vectors corresponding to eigenvalue -1
are $|01\rangle, |10\rangle$

Projection operator: $P_2 = |01\rangle\langle 01| + |10\rangle\langle 10|$

$$P_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\lambda_2 = -1}$$

8. $\Rightarrow A$ and B are unitarily diagonalizable.

$$\text{Let } A = U \lambda_A U^\dagger$$

$$, B = V \lambda_B V^\dagger$$

U, V are unitary; λ_A, λ_B are diagonal.

$$\Rightarrow A \otimes B = (U \lambda_A U^\dagger) \otimes (V \lambda_B V^\dagger)$$

$$= (U \otimes V) (\lambda_A \otimes \lambda_B) (U^\dagger \otimes V^\dagger)$$

$\Rightarrow U \otimes V$ is unitary and $\lambda_A \otimes \lambda_B$ is diagonal

$\Rightarrow A \otimes B$ is unitarily diagonalizable

$\Rightarrow$ Let λ_i be eigenvalues of A with eigenvector u_i and μ_j be eigenvalues of B with eigenvector v_j

$$\begin{aligned} \Rightarrow (A \otimes B)(u_i \otimes v_j) &= (A u_i) \otimes (B v_j) \\ &= (\lambda_i u_i) \otimes (\mu_j v_j) = \lambda_i \mu_j (u_i \otimes v_j) \end{aligned}$$

$\Rightarrow$ Eigenvalues of $A \otimes B$ are $\lambda_i \mu_j$ with corresponding eigenvectors $u_i \otimes v_j$.

$\Rightarrow$ So, eigenvalues of $A \otimes B$ are

$$\{ \lambda_i \mu_j : \lambda_i \text{ is eigenvalue of } A, \mu_j \text{ is eigenvalue of } B \}$$

with eigenvectors $u_i \otimes v_j$ where $\{u_i\}, \{v_j\}$ are orthonormal eigenbases for A, B respectively.

9. $\Rightarrow$ Consider applying partial trace operation on an arbitrary $d_1 d_2 \times d_1 d_2$ matrix R (R need not be density operator)

(a) $\Rightarrow \{ |j\rangle \}_{j=0}^{d_2-1}$ — orthonormal basis of B

$\Rightarrow$ Kraus operators — $K_j = I_{d_1} \otimes \langle j|$, $j=0$ to d_2-1

Map $\mathcal{H}_A \otimes \mathcal{H}_B \longrightarrow \mathcal{H}_A$

$$\Rightarrow \text{Tr}_B(R) = \sum_{j=0}^{d_2-1} K_j R K_j^\dagger = \sum_{j=0}^{d_2-1} (I_{d_1} \otimes \langle j|) R (I_{d_1} \otimes |j\rangle)$$

Proof:

$$\text{By definition, } \text{Tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2| \cdot (\langle b_2|b_1\rangle)$$

Define map $\phi(R) = \sum_j (\mathbb{I} \otimes \langle j|) R (\mathbb{I} \otimes |j\rangle)$

Keep $R = |a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|$

$$\phi(R) = \sum_{j=0}^{d_2-1} (\mathbb{I} \otimes \langle j|) (|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) (\mathbb{I} \otimes |j\rangle)$$

$$= \sum_{j=0}^{d_2-1} (\mathbb{I} |a_1\rangle\langle a_2| \mathbb{I}) \otimes (\langle j|b_1\rangle\langle b_2|j\rangle)$$

$$= \sum_{j=0}^{d_2-1} (\langle j|b_1\rangle\langle b_2|j\rangle) (|a_1\rangle\langle a_2|)$$

$$= |a_1\rangle\langle a_2| \left(\sum_{j=0}^{d_2-1} \langle j|b_1\rangle\langle b_2|j\rangle \right)$$

Now we use completeness condition

$$\sum_j \langle j|b_1\rangle\langle b_2|j\rangle = \sum_j \langle b_2|j\rangle\langle j|b_1\rangle$$

$$= \langle b_2| \underbrace{\left(\sum_j |j\rangle\langle j| \right)}_{\mathbb{I}} |b_1\rangle = \langle b_2|b_1\rangle$$

$$\phi(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2| \cdot \langle b_2|b_1\rangle$$

$$\phi(R) = \sum_j K_j R K_j^\dagger \quad ; \quad K_j = \mathbb{I} \otimes \langle j|$$

as per our original definition

$$\text{and } \phi(R) = \text{Tr}_B(R)$$

$$\Rightarrow [\text{Tr}_B(R)]_{a,a'} = \sum_{j=0}^{d_2-1} R(a,j), (a',j)$$

$$\Rightarrow \boxed{\text{Tr}_B(R) = \sum_{j=0}^{d_2-1} (\mathbb{I}_{d_1} \otimes \langle j|) R (\mathbb{I}_{d_1} \otimes |j\rangle)}$$

$\Rightarrow$ Completeness condition: $\hookrightarrow$ shorthand: $\sum_{j=0}^{d_2-1} \langle j|_B R |j\rangle_B$

$$\sum_{j=0}^{d_2-1} K_j^\dagger K_j = \sum_{j=0}^{d_2-1} (\mathbb{I}_{d_1} \otimes \langle j|) (\mathbb{I}_{d_1} \otimes |j\rangle)$$

$$= \sum_{j=0}^{d_2-1} (\mathbb{I}_{d_1} \otimes \mathbb{I}_{d_1}) \otimes (|j\rangle \langle j|)$$

$$= \sum_{j=0}^{d_2-1} \mathbb{I}_{d_1} \otimes |j\rangle \langle j| = \mathbb{I}_{d_1} \otimes \sum_{j=0}^{d_2-1} |j\rangle \langle j|$$

$$= \mathbb{I}_{d_1} \otimes \mathbb{I}_{d_2} = \mathbb{I}_{d_1 d_2}$$

$$\Rightarrow \boxed{\sum_{j=0}^{d_2-1} K_j^\dagger K_j = \mathbb{I}_{d_1 d_2}} \rightarrow \text{Completeness condition holds}$$

(b) $\Rightarrow$ By definition, $\text{Tr}_A(|a_1\rangle \langle a_2| \otimes |b_1\rangle \langle b_2|)$
 $= \langle a_2 | a_1 \rangle |b_1\rangle \langle b_2|$

We check linearity to all operators on $\mathcal{H}_A \otimes \mathcal{H}_B$

$\Rightarrow$ let $\{|i\rangle\}_{i=0}^{d_1-1}$ is orthonormal basis of $\mathcal{H}_A$

$$\Rightarrow \text{let } \psi(R) = \sum_{i=0}^{d_1-1} (\langle i| \otimes \mathbb{I}_B) R (|i\rangle \otimes \mathbb{I}_B)$$

Kraus operators $\longrightarrow K_i = \langle i| \otimes \mathbb{I}_B$

Map from $\mathcal{H}_A \otimes \mathcal{H}_B \longrightarrow \mathcal{H}_B$

We need to prove $\psi(R) = \text{Tr}_A(R)$

$$\Rightarrow \text{Let } R = |a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|$$

$$\begin{aligned} \psi(R) &= \sum_{i=0}^{d_1-1} (\langle i| \otimes I_B) (|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) (|i\rangle \otimes I_B) \\ &= \sum_{i=0}^{d_1-1} (\langle i|a_1\rangle\langle a_2|i\rangle) \otimes (I_B |b_1\rangle\langle b_2| I_B) \\ &= \sum_{i=0}^{d_1-1} (\langle i|a_1\rangle\langle a_2|i\rangle) |b_1\rangle\langle b_2| \\ &= |b_1\rangle\langle b_2| \sum_{i=0}^{d_1-1} \langle i|a_1\rangle\langle a_2|i\rangle \\ &= |b_1\rangle\langle b_2| \sum_{i=0}^{d_1-1} \langle a_2|i\rangle\langle i|a_1\rangle \\ &= |b_1\rangle\langle b_2| \langle a_2|a_1\rangle \underbrace{\left(\sum_{i=0}^{d_1-1} |i\rangle\langle i| \right)}_{I_{d_1} \text{ or } I_A} \end{aligned}$$

$$\psi(R) = \psi(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = \langle a_2|a_1\rangle |b_1\rangle\langle b_2|$$

by linearity

$$\psi(R) = \text{Tr}_A(R)$$

$$\Rightarrow \boxed{\text{Tr}_A(R) = \sum_{i=0}^{d_1-1} (\langle i| \otimes I_B) R (|i\rangle \otimes I_B)}$$

Kraus form $\rightarrow K_i = \langle i| \otimes I_B$ is Kraus operator

$$\boxed{\text{Tr}_A(R) = \sum_{i=0}^{d_1-1} K_i R K_i^\dagger}$$

$\Rightarrow$ Completeness condition:

$$\sum_{i=0}^{d_1-1} K_i^\dagger K_i = \sum_{i=0}^{d_1-1} (|i\rangle \otimes I_B) (\langle i| \otimes I_B)$$

$$= \sum_{i=0}^{d_1-1} (|i\rangle \langle i|) \otimes (I_B I_B)$$

$$= \left(\sum_{i=0}^{d_1-1} |i\rangle \langle i| \right) \otimes I_{d_2} = I_{d_1} \otimes I_{d_2} = I_{d_1 d_2}$$

$$\Rightarrow \boxed{\sum_{i=0}^{d_1-1} K_i^\dagger K_i = I_{d_1 d_2}} \rightarrow \text{Completeness condition holds}$$

10. $\Rightarrow$ Pauli operators:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

(a) $\Rightarrow$ let $A = \begin{bmatrix} a & b \\ \bar{b} & d \end{bmatrix}$ be Hermitian i.e. $A^\dagger = A$.
 $a \in \mathbb{R}, d \in \mathbb{R}, b = x + iy, \bar{b} = x - iy, x \in \mathbb{R}, y \in \mathbb{R}$

$\Rightarrow$ we want real coefficients $\alpha, \beta, \gamma, \delta$ such that

$$A = \alpha I + \beta X + \gamma Y + \delta Z$$

$$\Rightarrow \alpha I + \beta X + \gamma Y + \delta Z = \begin{bmatrix} \alpha + \delta & \beta + i\gamma \\ \beta - i\gamma & \alpha - \delta \end{bmatrix}$$

$$\Rightarrow \alpha = \frac{a+d}{2}, \delta = \frac{a-d}{2}, \beta = x, \gamma = y.$$

All these coefficients are real.

$\Rightarrow$ So, every 2×2 Hermitian matrix is a real linear combination of Pauli operators.

(b) $\Rightarrow$ Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow a, b, c, d \in \mathbb{C}$

$\Rightarrow$ Let $\alpha, \beta, \gamma, \delta \in \mathbb{C}$ such that

$$M = \alpha I + \beta X + \gamma Y + \delta Z$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \alpha + \delta & \beta + i\gamma \\ \beta - i\gamma & \alpha - \delta \end{bmatrix}$$

$$\alpha = \frac{a+d}{2}, \quad \delta = \frac{a-d}{2}, \quad \beta = \frac{b+c}{2}, \quad \gamma = \frac{b-c}{2i}$$

$\Rightarrow$ Every 2×2 complex matrix is a complex linear combination of I, X, Y, Z .

$\Rightarrow$ Let $d_1 I + d_2 X + d_3 Y + d_4 Z = 0$

$$\begin{bmatrix} d_1 + d_4 & d_2 + i d_3 \\ d_2 - i d_3 & d_1 - d_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$d_1 = d_4 = 0, \quad d_2 = d_3 = 0.$$

So, if $d_1 I + d_2 X + d_3 Y + d_4 Z = 0$ then

$$d_i = 0 \quad \forall i = 1, 2, 3, 4.$$

$\Rightarrow$ These operators form a basis for $\mathbb{C}^{2 \times 2}$.

(c) $\Rightarrow$ n -qubit Paulis are the operators

$$\{A_1 \otimes \dots \otimes A_n : A_i \in \{I, X, Y, Z\}, i = 1 \text{ to } n\}$$

$\Rightarrow$ ~~let~~ 4^n possible operators.

$\Rightarrow \mathbb{C}^{2^n \times 2^n}$ has dimension $(2^n)^2 = 4^n$. — ①

$\Rightarrow$ So, we need to show that n -qubit Pauli operators are linearly independent

$\Rightarrow$ ~~let~~ $\langle M, N \rangle = \text{Tr}(M^\dagger N)$ — Hilbert Schmidt inner product

$\Rightarrow$ For single qubit Pauli operators $P, Q \in \{I, X, Y, Z\}$

$$\text{Tr}(PQ) = \begin{cases} 2 & \text{if } P = Q \\ 0 & \text{if } P \neq Q \end{cases}$$

$\Rightarrow$ Take 2 n -~~qubit~~ qubit Pauli operators ~~M, N~~ P_i, P_j

$$\begin{aligned} \langle P_i, P_j \rangle &= \text{Tr}(P_i^\dagger P_j) \\ &= \text{Tr}((A_1 \otimes A_2 \otimes \dots \otimes A_n)^\dagger (B_1 \otimes \dots \otimes B_n)) \\ &= \prod_{j=1}^n \text{Tr}(A_j B_j) ; A_j, B_j \in \{I, X, Y, Z\} \end{aligned}$$

$$\Rightarrow \langle P_i, P_j \rangle = 2^n \delta_{ij}$$

$\Rightarrow$ n -qubit Pauli operators are mutually orthogonal w.r.t inner product.

$\Rightarrow$ Let $\sum_i c_i P_i = 0$.

$$0 = \langle P_j, 0 \rangle = \sum_i c_i \langle P_j, P_i \rangle = 2^n c_j$$

$$c_j = 0 \quad \forall j.$$

$\Rightarrow$ n -qubit Pauli operators are linearly independent (2)

$\Rightarrow$ So, n -qubit Pauli operators form a basis for $\mathbb{C}^{2^n \times 2^n}$ by statements (1) and (2)

THE END